Name _____ Date _____

Standards-Based Practice

Expressions and Equations:
Write and Graph Inequalities

Write an inequality of the form $x > c$ or $x < c$ to represent a constraint or condition in a real-world or mathematical problem. Recognize that inequalities of the form $x > c$ or $x < c$ have infinitely many solutions; represent solutions of such inequalities on number line diagrams.

1. Graph the inequality on a number line.

$x > 5$

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2. Graph the inequality on a number line.

$7 \geq x$

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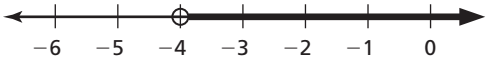
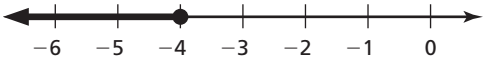
3. Match the inequality with its graph.

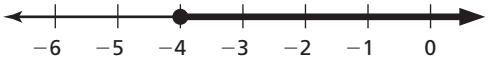
$x \geq -4$



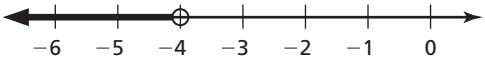
$x < -4$



$x > -4$



$x \leq -4$



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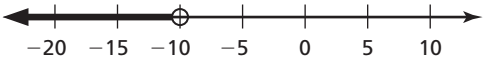
4. Which inequality represents the graph?



A. $x > -10$
B. $x < -10$
C. $x \leq -10$
D. $x \geq -10$

Expressions and Equations

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5. The boiling point of water is 212°F. Write and graph an inequality to represent the temperatures at which water will boil.

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6. When catching Dungeness Crab, the maximum number of crabs you can keep is 10 per day. To be able to keep a crab, the shell must be at least 5.75 inches across.

   a. Write and graph an inequality that represents the number of crabs you are allowed to keep per day.

   b. Write and graph an inequality that represents the lengths of crabs you are allowed to keep.

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7. You always keep more than $2,000 in your savings account. Write and graph an inequality to represent the amount of money that is in your savings account.

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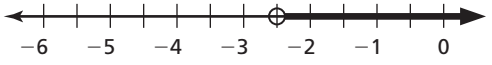
8. Write an inequality that represents the graph.



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